MAIS PAC

IPhO 02.05: Hydrodynamics and Surface Tension · Lecture + Problems

Hydrodynamics and Surface Tension

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Pressure and its consequences

Pressure is force per unit area, and in a fluid at rest it acts equally in every direction — a scalar, not a vector. Two results follow immediately:

- **Depth dependence.** $p = p_0 + \rho g h$. The shape of the container is irrelevant, only the depth.
- **Pascal's principle.** A pressure change applied at one point in an enclosed fluid is transmitted undiminished everywhere. That is the hydraulic press, and it trades force against distance, never energy.

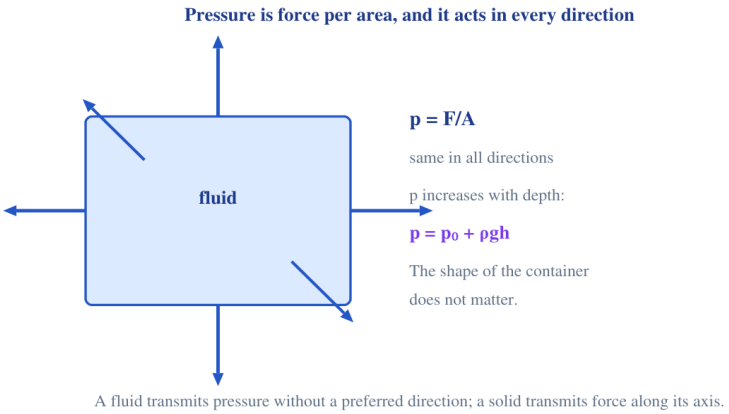


Figure 1. Pascal's principle follows from this isotropy: a pressure change applied at one point is transmitted undiminished throughout an enclosed fluid.

2. Buoyancy from a pressure difference

The pressure on the bottom of a submerged block exceeds that on its top by ρ_f g h, so the net upward force is

$$F_b = \rho_f V g, \tag{1}$$

the weight of displaced fluid. That is Archimedes' principle, derived rather than asserted.

Three consequences are worth remembering:

- A body floats when its mean density is less than the fluid's, and it displaces its own weight.
- A body in equilibrium fully submerged feels an apparent weight reduced by F_b; this is the classical method of measuring density.
- A floating object's **stability** depends on the position of the metacentre, not on the centre of mass alone — the reason a wide boat is stable and a narrow one capsizes.

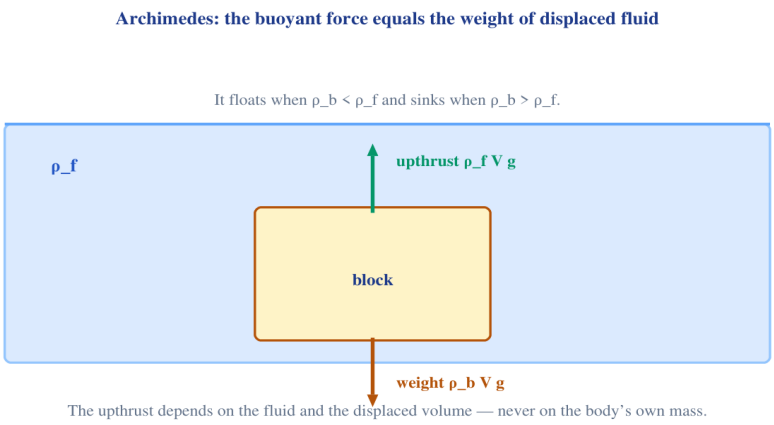


Figure 4. Archimedes' principle is a pressure-integration result: the pressure at the bottom of the block exceeds that at the top, and the difference is the weight of displaced fluid.

3. Continuity and Bernoulli

For a steady, incompressible flow, mass conservation gives the **continuity equation**:

$$A_1 v_1 = A_2 v_2. \tag{2}$$

Along a streamline, energy conservation per unit volume gives **Bernoulli's equation**:

$$p + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}. \tag{3}$$

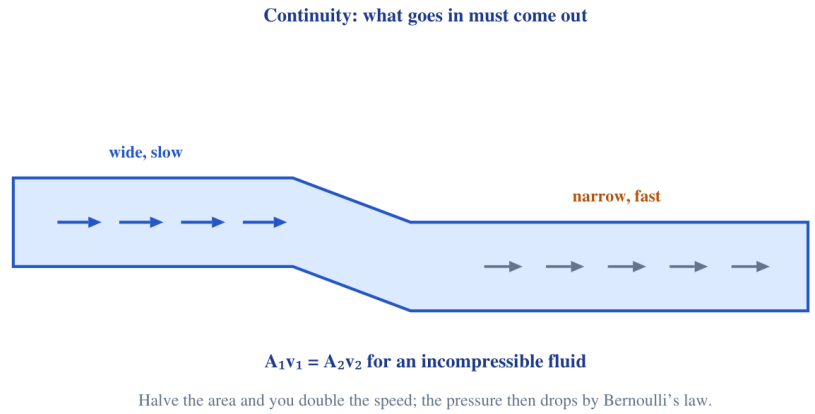


Figure 2. Continuity is mass conservation for an incompressible fluid. It is why a river runs faster in a narrow gorge than over a wide floodplain.

The three terms are pressure energy, kinetic energy and gravitational potential energy per unit volume. Reading Bernoulli as “fast flow means low pressure” is only safe when the pipe is level; when the pipe rises, the pressure **gains** ρ gh even if the flow slows.

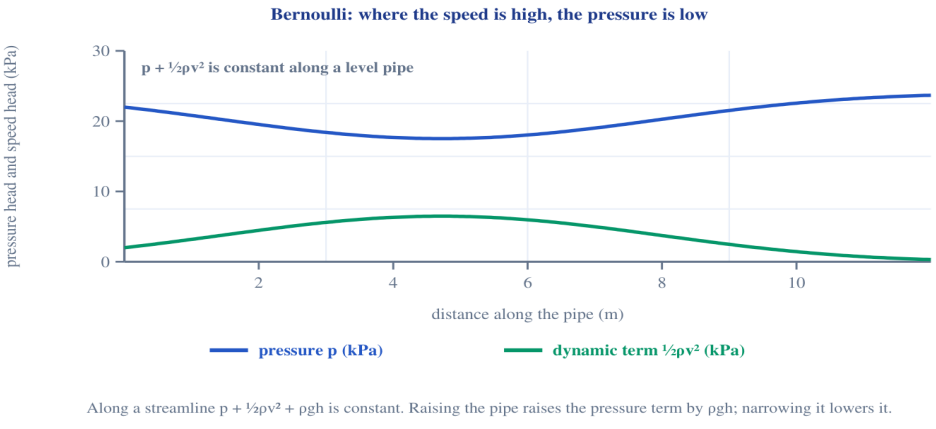


Figure. Along a streamline p + ½ρv² + ρgh is constant. Raising the pipe raises the pressure term by ρgh; narrowing it lowers it.

Two standard applications:

- **Torricelli's law.** For a small hole a depth h below the free surface, the speed is v = √(2gh) — the same as free fall from that height. The jet then follows projectile motion.
- **The Venturi effect.** A narrowing speeds the flow and lowers the pressure; this is how a carburettor, an atomiser and a Pitot-static airspeed indicator work.

Interactive experiment: A tank with a hole: Torricelli’s law and the jet — open the online handout to use it.

4. Surface tension and capillarity

Surface tension γ is energy per unit area, or equivalently force per unit length along a surface. It explains why:

- **Small drops are spherical** — a sphere has the smallest area for a given volume.
- **A curved surface sustains a pressure difference:**

$$\Delta p = (2\gamma)/(r) \{(\text{one surface})\}, \Delta p = (4\gamma)/(r) \{(\text{a soap bubble, two surfaces})\}. \quad (4)$$

- **Liquid rises in a narrow tube:**

$$h = (2\gamma\cos\theta)/(\rho g r), \quad (5)$$

so a narrow tube lifts water higher, and a non-wetting liquid (mercury, $\theta > 90^\circ$) is **depressed**. - **Small objects float** — a needle on water is supported by the deformed surface, not by buoyancy.

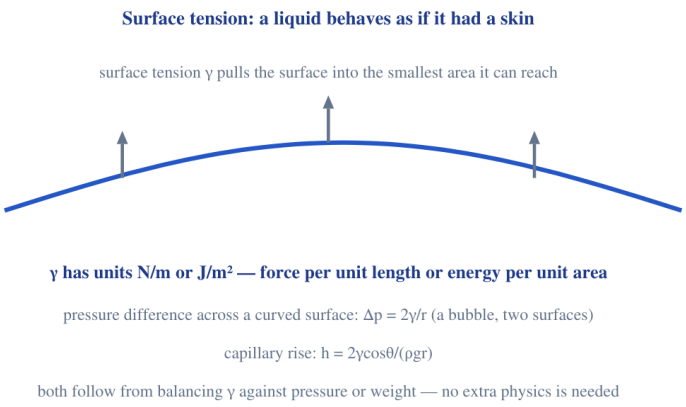


Figure 5. Small objects can float on water not because they are light but because deforming the surface costs energy $\gamma\Delta A$.

5. Real fluids: viscosity and Reynolds number

Real fluids resist flow. For slow flow through a pipe the volume rate is given by Poiseuille's law, $Q \propto r^4,\Delta p/(L)$ — the fourth power of the radius is why arteries narrow dangerously. Whether a flow is smooth (laminar) or turbulent is set by the **Reynolds number**

$$\{Re\} = (\rho v L)/(), \quad (6)$$

with a transition near $\{Re\}2000$ for pipe flow. For a sphere at low Reynolds number, Stokes' law gives drag $F = 6\pi r v$, which produces the terminal velocity $v_T = (2r^2(\rho_s-\rho_f)g)/(9)$ — a result that scales as r^2 and explains why fine dust settles slowly.

The fluid checklist. Decide first whether the fluid is incompressible (all liquids, gases at low speed) → write continuity → write Bernoulli between two points **on the same streamline** → use $p = p_0+\rho gh$ for the static part → check the limits (zero speed, zero hole size, zero gravity) → only then consider viscosity or turbulence.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Pressure in the deep.

Find **(a)** the gauge pressure at 10 {km} depth in the ocean (ρ = 1030 {kg,m^(-3)}). (b) Find the total force on a circular porthole of radius 0.15 {m} at that depth. (c) Explain why the total pressure, not the gauge pressure, determines whether the glass fails.

Worked solution →

[1] Problem 2 — Floating and sinking. A block of wood (ρ = 600 {kg,m^(-3)}) floats in water. (a) Find the fraction of its volume that is submerged. (b) Find the fraction submerged when it floats in ethanol (ρ = 789 {kg,m^(-3)}). (c) Show that the answer depends only on the density ratio, and explain what happens in a liquid of exactly equal density.

Worked solution →

[2] Problem 3 — Torricelli and the jet. A tank of cross-section A holds water to depth h above a small hole of area a at height H above the ground. (a) Derive v=√(2gh) for the efflux speed, stating which approximation you use. (b) Derive the horizontal range of the jet. (c) Find the hole height that maximises the range for a fixed h + H, and discuss how the approximation fails as a→ A.

Worked solution →

[2] Problem 4 — The Venturi meter. A horizontal pipe of area A_1 narrows to A_2 and then returns to A_1. The pressure difference between the wide and narrow sections is measured as Δ p. (a) Find the volume flow rate in terms of A_1, A_2, Δ p and ρ. (b) Show that the result for A_2→ A_1 and for A_2→0 behaves sensibly. (c) A Venturi meter reads Δ p = 12 {kPa} for airflow (ρ = 1.2 {kg,m^(-3)}) with A_1 = 0.020 {m²} and A_2 = 0.008 {m²}: find Q.

Worked solution →

[2] Problem 5 — Capillary rise. Water rises 8.0 {cm} in a capillary tube. (a) Find the tube radius using γ = 0.072 {N,m^(-1)}, θ≈0. (b) Find the rise in a tube of half that radius. (c) Explain why mercury is depressed rather than raised, and why the **height** of the rise depends on the tube but the **pressure** under the meniscus depends on its curvature.

Worked solution →

[2] Problem 6 — The hydraulic press. A hydraulic press has a small piston of diameter 2.0 {cm} and a large piston of diameter 20 {cm}. A force of 200 {N} is applied to the small piston. (a) Find the force on the large piston. (b) Find how far the small piston must move to lift the large one by 5.0 {cm}. (c) Show that the work done is the same on both sides and explain why the press does not violate energy conservation.

Worked solution →

[2] Problem 7 — The floating ice cube. An ice cube (ρ = 917 {kg,m^(-3)}) floats in a glass of water. (a) Show that the water level does not change when the ice melts. (b) Work out what happens to the level if the ice encloses a bubble of air — or a stone. (c) Explain the answer in terms of displaced water rather than weights.

Worked solution →

[2] Problem 8 — Bernoulli in a siphon. A siphon is a tube over a wall, with its intake at height h_1 below the free surface of a container and its outlet at a greater depth h_2. (a) Show that the efflux speed at the outlet is √(2g(h_2-h_1)) if the outlet is the lowest point. (b) Find the pressure at the highest point of the siphon and explain why a siphon cannot lift water more than about 10 {m}. (c) Explain why a siphon stops working if a bubble of air enters the tube.

Worked solution →

[2] Problem 9 — Stokes' law and terminal velocity. A spherical bead of radius r and density ρ_s falls through a liquid of density ρ_f and viscosity . (a) Show that the terminal speed is v_T = (2r²(ρ_s-ρ_f)g)/(9). (b) Find the ratio of the terminal speeds of two beads with radii r and 2r. (c) Estimate the Reynolds number for a 1.0 {mm} sand grain in water (= 1.0×10^(-3) {Pa,s}, ρ_s = 2650 {kg,m^(-3)}) and comment on whether Stokes' law applies.

Worked solution →

[3] Problem 10 — A barge, a bridge and a load. A rectangular barge of length L = 20 {m}, width W = 5.0 {m} and mass M = 3.0×10^4 {kg} floats in fresh water. (a) Find its draft. (b) A 1.5×10^4 {kg} crane is loaded at one end: by how much does the barge tilt, and does it swamp? (c) A 5.0×10^4 {kg} container is lowered onto the barge from a bridge with a cable: describe how the tension in the cable and the barge's draft change as it is lowered, from the instant it touches the water to the instant it rests fully on the deck.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official

marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

HyperPhysics mechanics map

A compact concept map for revising definitions and standard results quickly.